

INTRODUCTION

- Alzheimer's disease (AD) is a neurodegenerative disorder characterized by memory loss and marked by changes in brain function.
- These changes are evident in resting-state functional network connectivity (FNC).
- Insomnia is common in AD, but it remains unclear how insomnia relates to brain connectivity, or whether that relationship is different in AD.
- Most studies have looked only at patients from memory clinics, so less is known about what these patterns look like in the general population.
- Here we used the UK Biobank, a large dataset from the general public, to compare FNC in people with and without AD, and in people with and without insomnia.

METHODS

- Participants: UK Biobank resting-state fMRI (imaging visit).
- Analysis sample: N = 2,827 (participants answering "sometimes" on the insomnia item were excluded).
- AD: 22 (11 with insomnia, 11 without). Non-AD: 2,805 (1,422 with insomnia, 1,383 without).
- Subject-level brain networks were identified through independent component analysis spatially constrained to the NeuroMark 2.2 template.
- The template is a standardized map of 105 intrinsic connectivity networks (ICNs) sorted into seven larger domains.
- FNC was estimated for each subject through a Pearson correlation of the time courses from these ICNs.
- A single GLM tested the effect of AD, the effect of insomnia, and the AD $\times$ insomnia interaction together, adjusting for age, sex, race, mean framewise displacement, and imaging site.
- Results were corrected for multiple comparisons using the false discovery rate (FDR).

ACKNOWLEDGMENTS

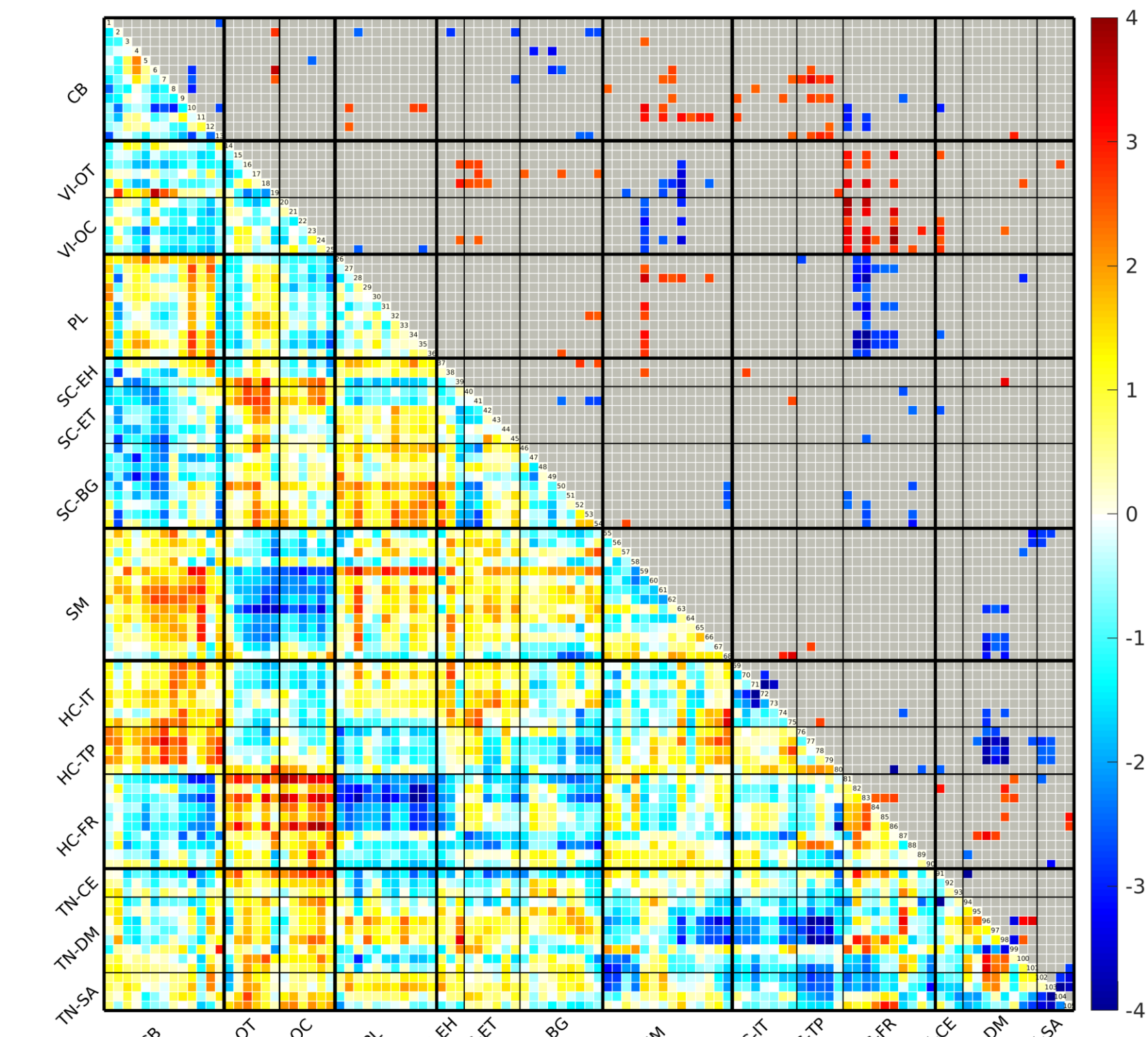
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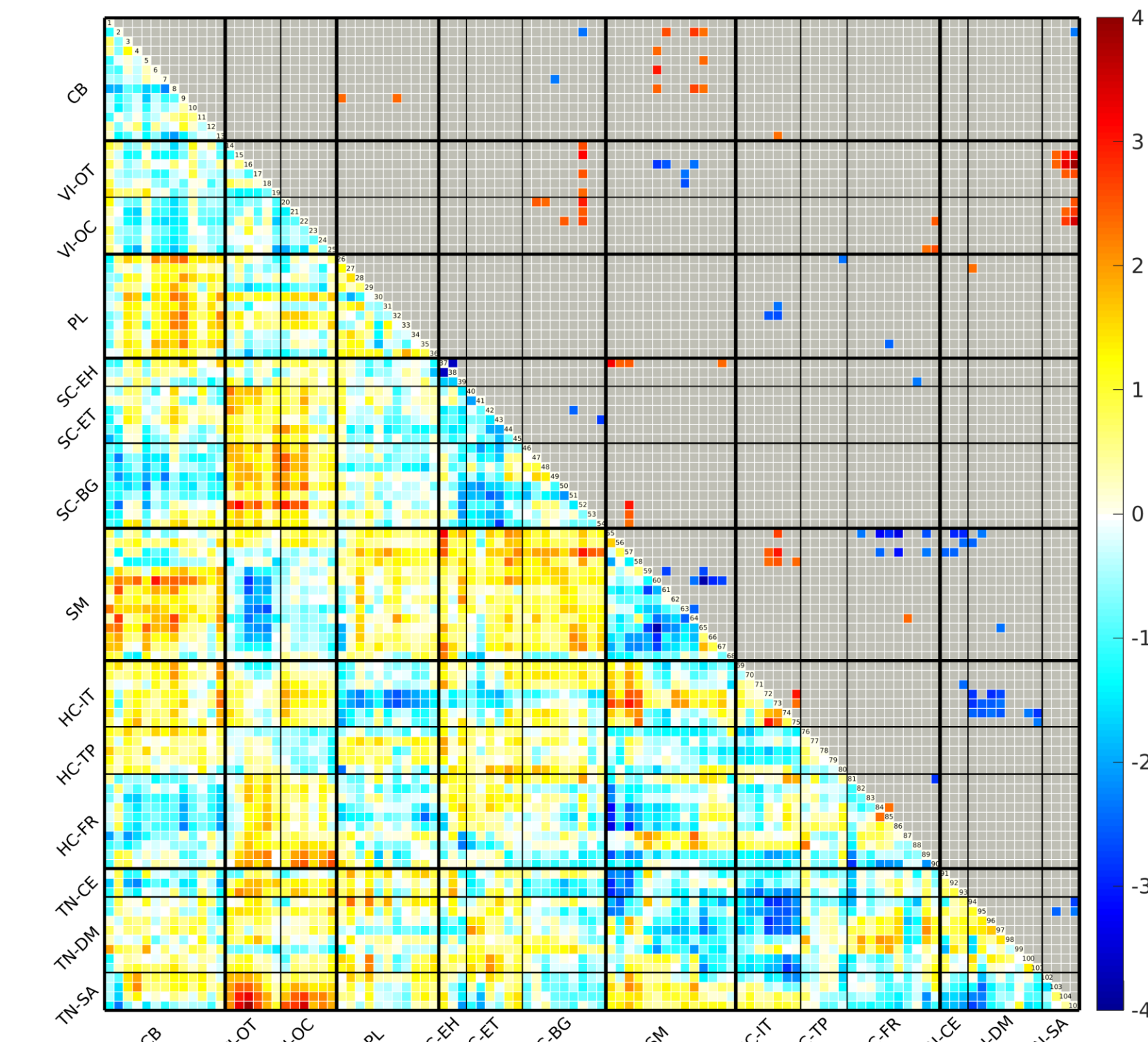
NeuroMark 2.2
reference

RESULTS



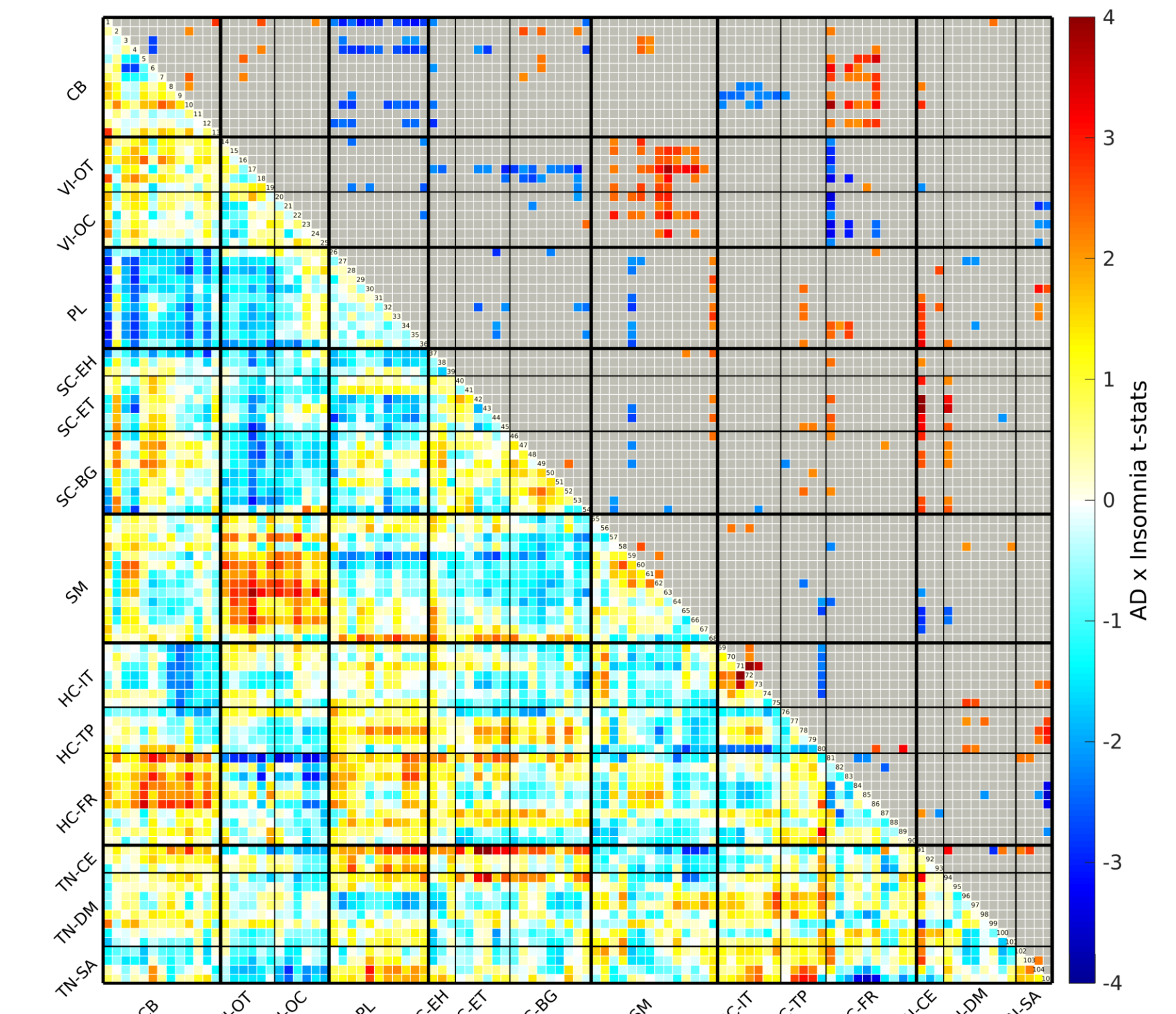
AD: widespread FNC differences

Figure 1 | Effect of AD. Upper triangle = connections surviving correction; lower triangle = all connections. Warm = FNC more positive in the AD group than the non-AD group; cool = more negative.



Insomnia: visual, sensorimotor, and salience differences

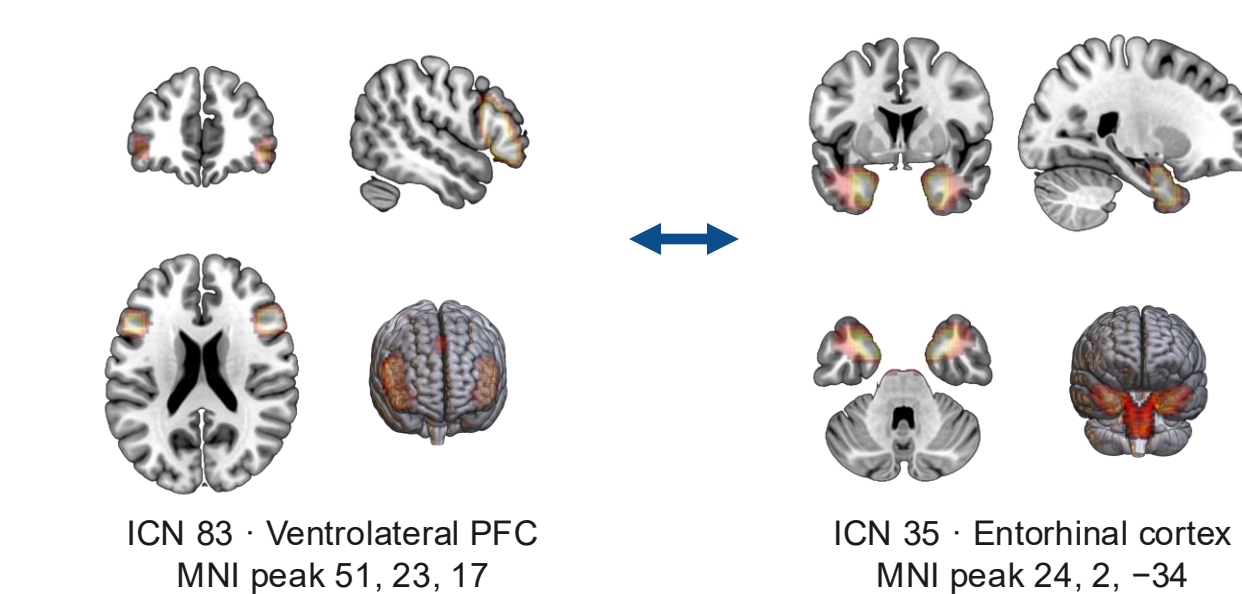
Figure 2 | Effect of insomnia, shown as in Figure 1. Warm = more positive in the insomnia group than the non-insomnia group; cool = more negative.



AD $\times$ Insomnia: strongest in visual-sensorimotor FNC

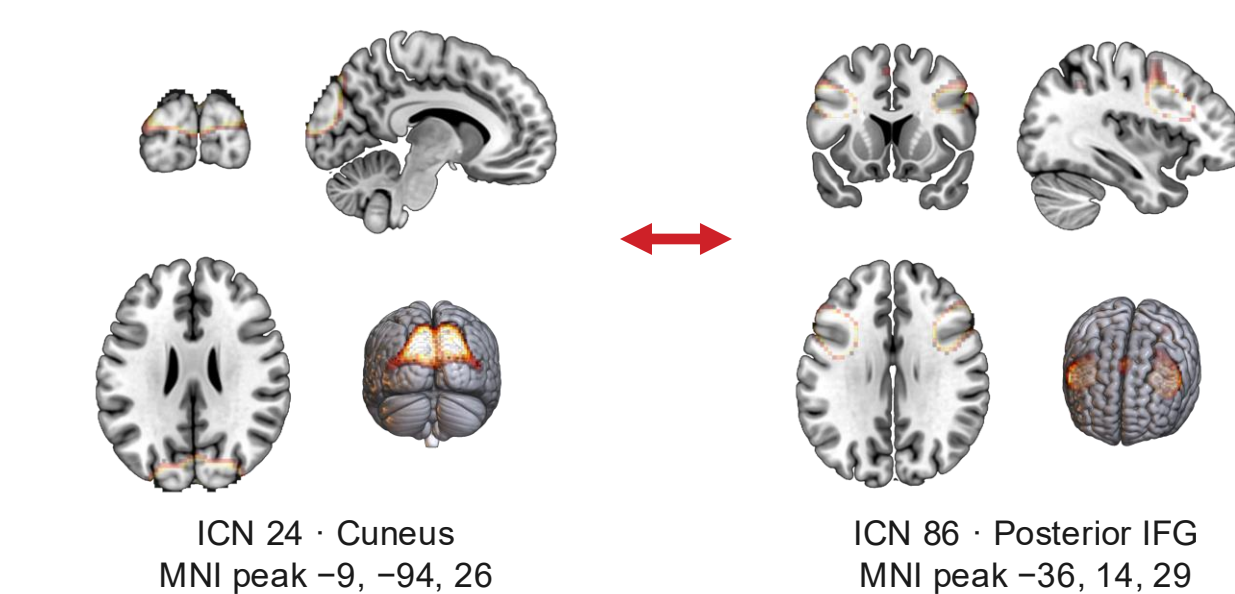
Figure 3 | AD $\times$ insomnia interaction, shown as in Figure 1. Warm = the effect of insomnia is more positive in AD than in non-AD; cool = more negative.

KEY NETWORK PAIRS



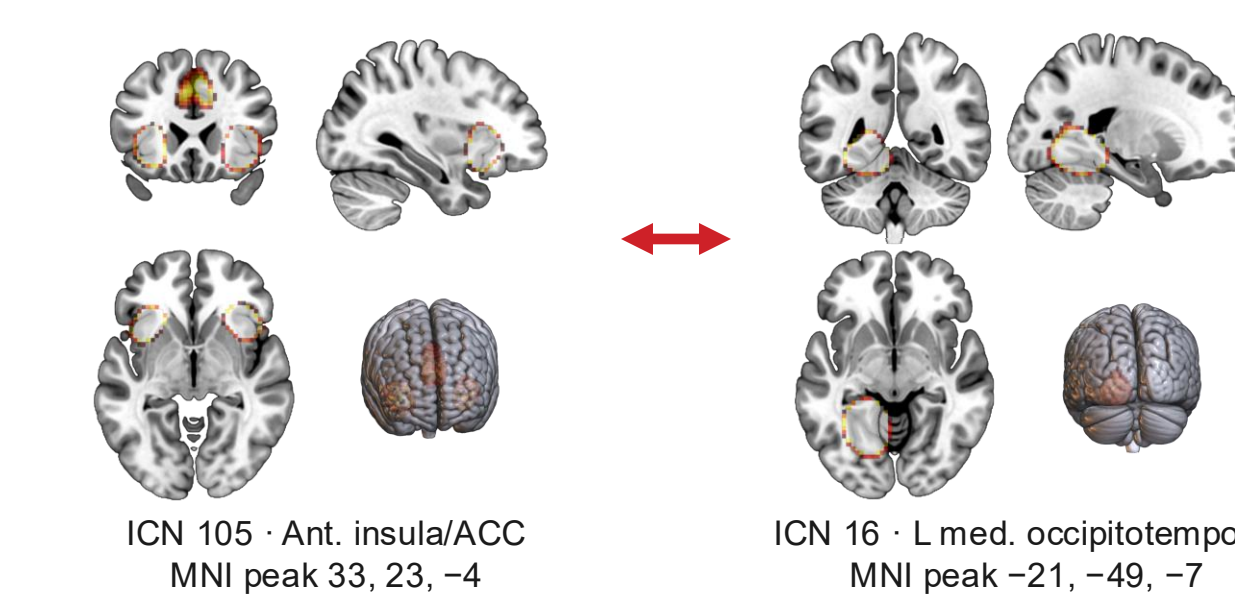
AD — Frontal $\leftrightarrow$ Paralimbic (more negative)

30 connections survived correction, all more negative in the AD group than the non-AD group.



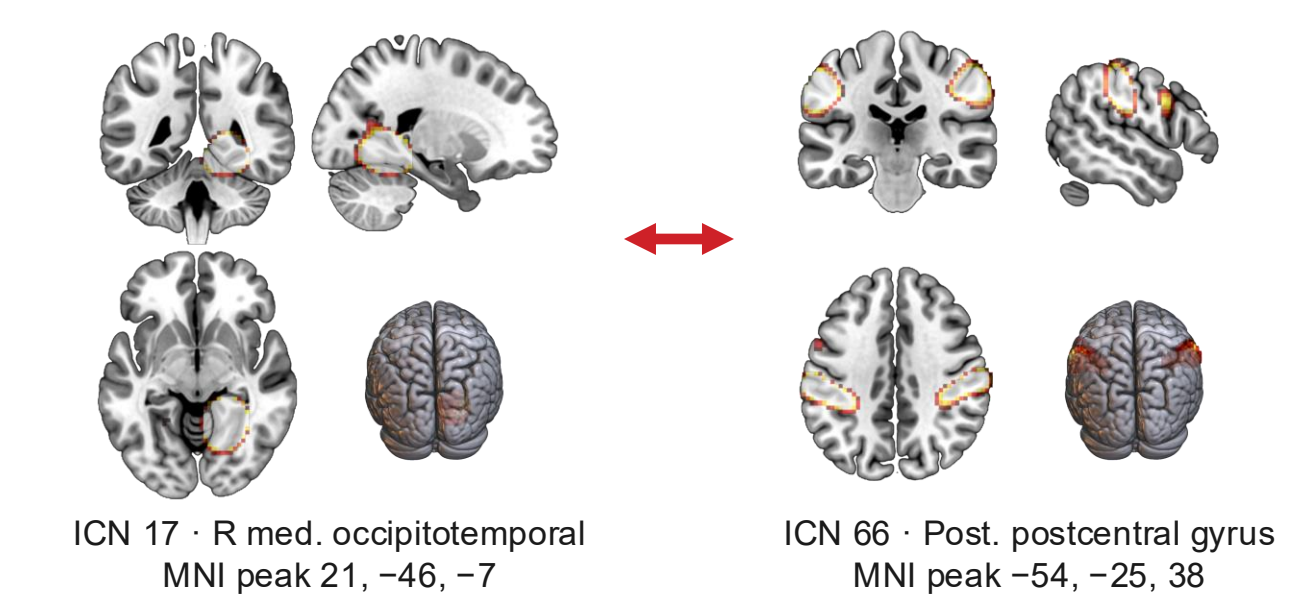
AD — Visual $\leftrightarrow$ Frontal (more positive)

19 connections survived correction, all more positive in the AD group than the non-AD group.



Insomnia — Salience $\leftrightarrow$ Visual (more positive)

13 connections survived correction, all more positive in the insomnia group than the non-insomnia group.



AD $\times$ Insomnia — Visual $\leftrightarrow$ Sensorimotor

29 connections survived correction; the effect of insomnia was more positive in AD than in non-AD.

DISCUSSION

- In AD, frontal-paralimbic FNC was more negative and visual-frontal more positive. Entorhinal cortex is affected early in AD, so weakened frontal-medial temporal coupling fits expected disease progression.
- Prior AD work emphasizes default-mode disruption; the prominence of visual networks here may reflect a general-population rather than a memory-clinic sample.
- Insomnia showed more positive salience-visual FNC. Salience and insular networks are commonly implicated in insomnia and hyperarousal, pointing to altered sensory gating.

- The interaction was strongest between visual and sensorimotor networks, so insomnia relates to connectivity differently depending on AD status rather than simply adding to it.
- Visual networks are the common thread, with different partners in each effect: frontal in AD, salience in insomnia, sensorimotor in the interaction.

Limitations & Next Steps

- Only 22 participants with AD (11 per insomnia cell); the interaction estimate rests on very small groups and needs replication.
- Insomnia was self-reported; objective sleep measures are needed.
- Unbalanced groups make imaging site and other covariates especially important.
- Longitudinal data could test whether these patterns precede diagnosis.